

Supplementary Material

Does Realism Predict Relevance? A Controlled Cross-Dataset Study of Generative Model Selection for Medical Image Synthesis

This supplement collects material referenced from the main paper: per-architecture diagrams for the five generative families (Section 1) and the complete two-modality PASD low-data augmentation sweep (Section 2), the generation-faithfulness check and the training configuration, and the transfer study abbreviated out of the main paper for space (Section 6).

1 Architecture diagrams

These diagrams summarize the generator and discriminator structure, conditioning signals, and training objective of each family exactly as implemented, with channel widths, block counts and latent dimensions as used in all experiments.

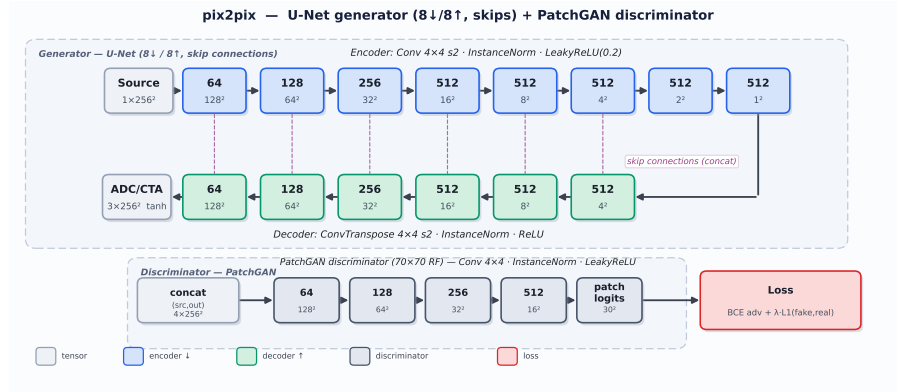


Figure 1: **pix2pix**. A U-Net generator (8 stride-2 down-sampling stages, $64 \rightarrow 512$ channels, mirrored decoder with skip-concatenation) maps the source modality to the target; a PatchGAN discriminator scores 70×70 patches of the (source, output) pair. Trained with BCE adversarial loss plus an L_1 reconstruction term.

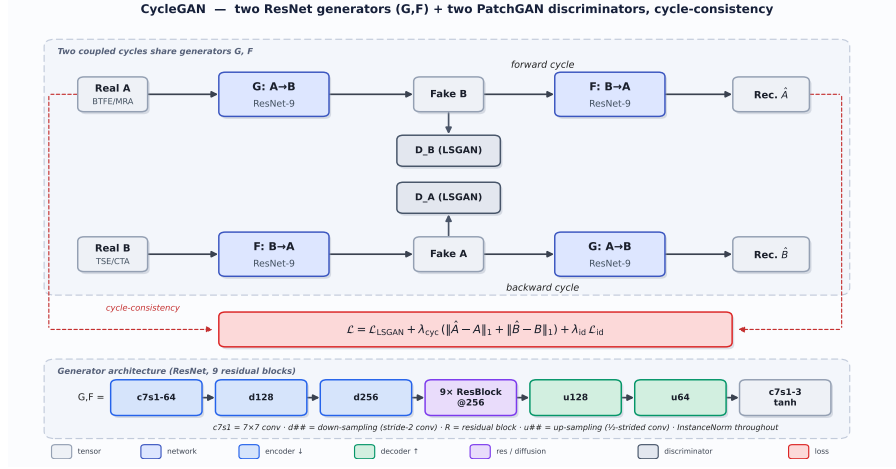


Figure 2: **CycleGAN**. Two ResNet generators ($G: A \rightarrow B$, $F: B \rightarrow A$, nine residual blocks each) and two PatchGAN discriminators are trained with an LSGAN adversarial loss, an L_1 cycle-consistency loss ($A \rightarrow B \rightarrow \hat{A}$ and back), and an identity term. An image buffer stabilizes the discriminator updates. Requires no paired supervision.

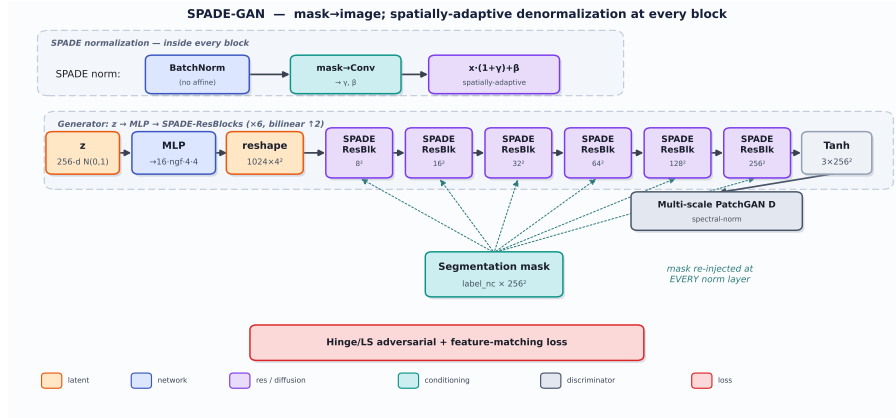


Figure 3: **SPADE-GAN**. A noise vector is projected and reshaped, then upsampled through six SPADE residual blocks to 256². The segmentation mask is re-injected at *every* normalization layer: a BatchNorm without affine parameters is modulated by spatially-varying (γ, β) predicted from the mask, intended to give tight mask adherence; in our PASD measurement it in fact adhered least tightly of the three generators scored (mask-Dice 0.807). A multi-scale PatchGAN provides adversarial supervision.

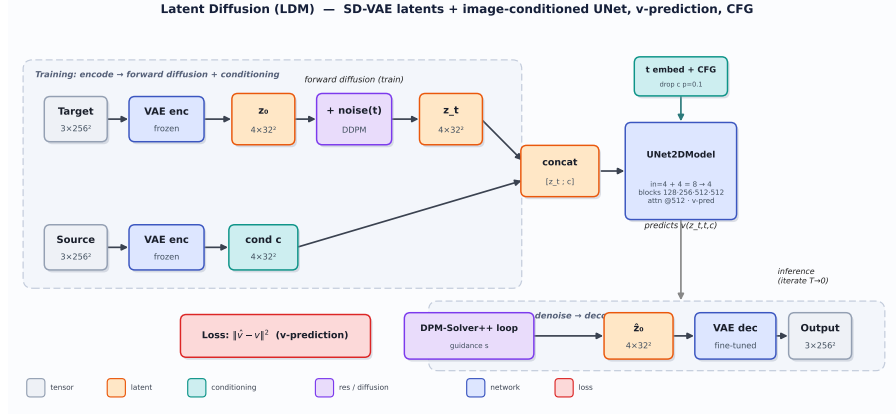


Figure 4: **Latent diffusion (LDM)**. A frozen Stable-Diffusion VAE encodes images to 4×32^2 latents. The target latent is noised (DDPM) and concatenated with the source-image latent (8 input channels); a UNet2DModel predicts the v -target with classifier-free guidance. Inference runs a DPM-Solver++ reverse loop, and the fine-tuned VAE decoder maps the denoised latent back to image space.

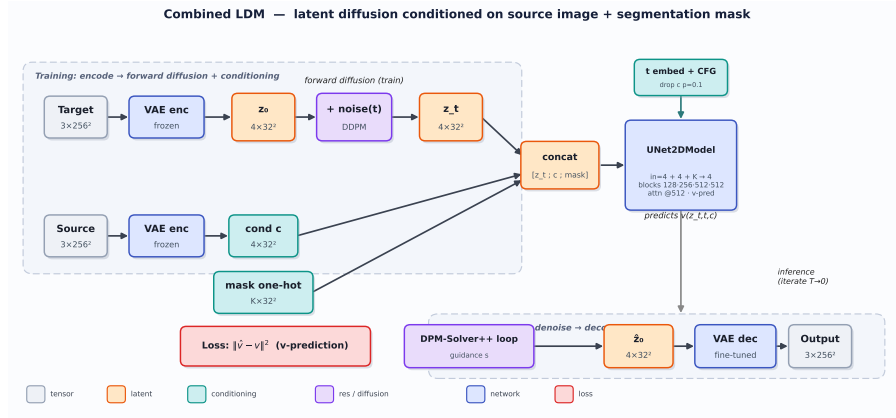


Figure 5: **Combined LDM**. As Figure 4, but the conditioning is the source-image latent *and* the one-hot segmentation mask at latent resolution (input channels $4 + 4 + K$). The extra mask channels inject task-relevant anatomy; this is the model that ranks first on downstream utility despite a worse FID than the plain LDM, by a margin that is positive but not statistically supported.

2 Complete PASD low-data augmentation sweep

Table 1 gives both modalities of the sweep abbreviated in the main paper. “xmod” denotes real data augmented with cross-modality CycleGAN pairs (BTfE $\leftrightarrow$ TSE). Cross-modality augmentation has the highest Dice at every training-set size on both modalities; every cell is a single run with no significance test, so this is a consistent direction rather than a measured effect.

How to read this table. N counts annotated axial *slices* of the target sequence, drawn at random from the full training split; it is not a patient count. Each augmented arm is paired with a synthetic set generated from the entire training annotation pool (the target sequence’s full training split for the mask $\rightarrow$ image arms, and the other sequence’s for +xmod: 2749 BTfE and 2941 TSE slices), irrespective of N , and the real-only arms were trained for more epochs (80, 60, 40, 30, 20 as N grows) than the augmented arms (20 throughout) to compensate for their smaller datasets. An augmented arm therefore differs from its real-only row in annotation count, image count and optimization budget at once, and the two columns must not be read as a data-efficiency comparison. The comparison between augmentation sources (+pix2pix, +LDM, +xmod) is matched and is the one discussed in the main paper. Every cell is a single run.

Table 1: PASD low-data augmentation sweep (TSTR placenta-Dice on the real test set, single seed per cell). N counts annotated training *slices*, not patients. Best per row in **bold**. “—” = condition not run.

N (real slices)	real	+pix2pix	+LDM	+xmod
<i>BTfE</i>				
50	0.806	0.864	0.865	0.879
100	0.835	0.874	0.876	0.886
200	0.848	0.880	0.879	0.892
500	0.880	0.881	0.884	0.899
full	0.900	0.897	0.897	0.907
<i>TSE</i>				
50	0.767	0.835	—	0.865
100	0.812	0.838	—	0.868
200	0.827	0.847	—	0.875
500	0.860	0.866	—	0.879
full	0.876	0.891	—	0.899

3 Generation faithfulness

Faithfulness asks whether a synthesized image respects its conditioning mask. Re-segmenting the synthetic PASD images against their input mask gives mask-Dice 0.926 (pix2pix), 0.911 (LDM) and 0.807 (SPADE). The two models that take the mask as an input channel adhere more tightly than SPADE, which drifts outside it; SPADE is also the weaker model overall (Section 5.4 of the main paper), so mechanism and quality are not separable.

4 Training configuration

Table 2: Training configuration (Section 4.2 of the main paper). All models train at 256×256 with mixed precision on one GPU. Diffusion models are trained stepwise, GANs epochwise.

Family	Optimizer	Batch	LR	Length
pix2pix	Adam $\beta_1=0.5$	8	2×10^{-4}	60 ep
CycleGAN	Adam $\beta_1=0.5$	4	2×10^{-4}	100 ep
SPADE	Adam (TTUR)	4	^a	100 ep
LDM	AdamW	16	1×10^{-4}	30k steps
Comb. LDM	AdamW	16	1×10^{-4}	30k steps
Segmenter	AdamW	16	2×10^{-4}	25 ep

Loss and schedule terms. pix2pix: L_1 weight 100. CycleGAN: cycle $\lambda_{\text{cyc}}=10$, identity $\lambda_{\text{id}}=5$, image buffer. SPADE: ^aseparate generator/discriminator rates $1 \times 10^{-4}/4 \times 10^{-4}$, feature-matching 10, perceptual (VGG) 10, EMA 0.999. Both diffusion models: cosine decay, EMA 0.9999, guidance dropout 0.1, 25 DPM-Solver++ steps, inference guidance 1.5. Segmenter: cosine decay, Dice+BCE loss. **Departures.** PASD SPADE uses batch 8, 200 epochs and no L_1 term; PASD LDM samples at guidance 3.0; both prostate diffusion models generate the downstream sets at guidance 1.0.

5 Selection framework

The design principles of Section 6 of the main paper, as a selection procedure.

Table 3: Selection framework. Both properties are measurable before training; the framework yields a prior, which DP2 requires be verified by TSTR.

Dataset property	Recommended choice	Verify
Paired, aligned modalities	Image-conditioned LDM; pix2pix baseline	TSTR vs. real
Same type, two sequences	Add CycleGAN cross-modality pairs	Gain at target N
Physically distinct modalities	Avoid unpaired translation as the sole source; it may still help as an augmenter	TSTR; cycle- L_1 cannot detect this
Mask coverage $\lesssim 7\%$ (all we tested)	Take the mask as an input channel, not through normalization alone	TSTR; mask-Dice of output
Coverage above $\sim 7\%$	Not tested	TSTR both ways
Goal is augmentation, real data ample	No source materially hurt; -0.4 to $+2.6$ pts	TSTR each source as an augmenter

6 Two recent methods do not transfer

Tuning the prostate T2→ADC LDM gave one strict improvement and one trade-off: lowering classifier-free guidance from 1.5 to 1.0 improved every metric, and ensemble-averaging four samples raised SSIM from 0.54 to 0.62 at an FID cost (72.7 to 88), exposed as a user-facing control. Neither of two recent published ideas transferred: adversarial latent diffusion in the spirit of SynDiff (Özbey et al., 2023) destabilized the backbone and regressed SSIM and FID (0.51 and 85, against 0.54 and 72.7), and a paired contrastive GAN in the spirit of CUT and AI-ADC (Park et al., 2020; Özyörük et al., 2024) raised SSIM to 0.58 but degraded FID to 128. Against AI-ADC’s reported 0.86 ROI-SSIM, cropping to the prostate-zone bounding box gives 0.54, matching our full-slice value, so the gap is real rather than a cropping artifact. These are negative results about *transfer*, not refutations: their scope is one dataset and our reimplementation, which we tuned less than the original authors did.